CS-413 Software Engineering
(Mid Semester Test I)

Date: September 16, 2025 **Time: 1 hr 15 Min.** **Marks: 20**

Answer all the questions. Each question carries 5 marks.

1. Compare and contrast the Spiral model and the Waterfall model in terms of risk management and validation activities. Under what conditions would one outperform the other?

2. In a complex software system, how can activity diagram be used to model and verify the concurrent and asynchronous workflow? Discuss how the use of Join and Fork nodes affect the control flow. Demonstrate with an example.

3. Draw Class Responsibility Collaborator (CRC) diagram for a "Loan Management System for a Bank". Identify and draw the suitable classes and their collaborator.

4. What is the Component and Connector (C&C) view in Software Architecture? How to represent C&C view for an 'Order Processing System' at an E-Commerce site.

School of Computer and Systems Sciences
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End Semester Examination,
Mansoon Semester 2025
CS 413—Software Engineering

Time: 3 Hrs. **Date: December 4, 2025** **Marks: 60**

Note: Answer all the questions. Each question carries (6+6=12) marks.

1. **(a)** The Agile philosophy emphasizes "customer collaboration over contract negotiation". How is this principle implemented in practice within the EXTREME process framework, and what is the role of the customer (Product Owner) in ensuring project success?

   (b) Explain the purpose of a Use Case Model in requirements engineering. For a Hospital Management System, explain how you would decompose a broad use case like "Manage Patient Records" into smaller, more precise use cases.

2. **(a)** Define and differentiate between the concepts of coupling and cohesion in software design. Explain the ideal relationship for both metrics (i.e., high/low) and why this balance is considered a cornerstone of good software architecture.

   (b) For an Online Shopping System, describe what the Context Diagram (Level 0) would include and what details must be excluded. Draw the Level 1 Data Flow Diagram of one of the major functionality of an Online Shopping System.

3. (a) What is Software refactoring? Consider a case of basic School Management System and explain the refactoring concepts through class diagram (I) before refactoring (II) after refactoring is done.

   (b) Differentiate clearly between the four main types of software maintenance: Corrective, Adaptive, Perfective, and Preventive. Provide a specific example of an activity for each type within the context of a deployed enterprise resource planning (ERP) system.

4. (a) Draw the Cause-Effect graph for a given scenario and discuss the possible test cases.

A program accepts three numerical inputs, representing the lengths of the sides of a triangle. The program's task is to determine if these inputs form a valid triangle and, if so, to classify its type:

- *Equilateral: All three sides are equal.*
- *Isosceles: Exactly two sides are equal.*
- *Scalene: No sides are equal.*

- *Not a Triangle: The given side lengths cannot form a valid triangle (e.g., if the sum of any two sides is less than or equal to the third side, or if any side is non-positive).*

(b) Draw the control flow diagram and possible test paths for the following piece of code.

```
while(x<100)
{ if(a[x]%2 == 0)
   { parity= 0; }
  else
   { parity= 1; }
  switch(parity)
   {
      case 0: println("a["+i+"] is even");
      case 1: println("a["+i+"] is odd");
      default: println("unexpected error");
   }
   x++;
}
p=true;
```

5. (a) What is the basic idea behind the COCOMO estimation model? Explain the meaning of effort adjustment factors (EAF) in COCOMO Model and how they influence final effort estimation.

(b) What is reverse engineering? Give an example of how it helps in understanding legacy systems. How does forward engineering fit into the reengineering life cycle?